Chapter 01 Part 2: The Periodic Table and Trends

1. The effective nuclear charge experienced by the outermost electron of Na is different than the effective nuclear charge experienced by the outermost electron of Ne. This difference best accounts for which of the following?

(A) Na has a greater density at standard conditions than Ne.

(B) Na has a lower first ionization energy than Ne.

(C) Na has a higher melting point than Ne.

(D) Na has a higher neutron-to-proton ratio than Ne.

(E) Na has fewer naturally occurring isotopes than Ne.

2. Which of the following ground-state electron configurations represents the atom that has the lowest first-ionization energy?

(A) $1s^2\ 2s^1$

(B) $1s^2\ 2s^2 2p^2$

(C) $1s^2\ 2s^2 2p^6$

(D) $1s^2\ 2s^2 2p^6\ 3s^1$

3.

Cl^-	Ar	K^+	Ca^{2+}

The four species above have the same electron configuration, $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$. Which of the following statements correctly identifies the species with the largest radius and provides an explanation based on Coulomb's law?

(A) Cl^-, because its nuclear charge exerts the least attractive force on the electrons in the $3p$ sublevel.

(B) Ar, because it is a neutral atom and the net force on the electrons in the $3p$ sublevel is zero.

(C) K^+, because the loss of an electron results in a smaller attractive force between the nucleus and the electrons in the $3p$ sublevel.

(D) Ca^{2+}, because the large positive charge on a cation causes increased repulsion among the electrons in the $3p$ sublevel.

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4.

Element	First Ionization Energy (kJ/mol)	Atomic Radius (pm)
B	801	85
C	1086	77
N	1400	75
O	1314	73
F	1680	72
Ne	2080	70

The table above shows the first ionization energy and atomic radius of several elements. Which of the following best helps to explain the deviation of the first ionization energy of oxygen from the overall trend?

(A) The atomic radius of oxygen is greater than the atomic radius of fluorine.

(B) The atomic radius of oxygen is less than the atomic radius of nitrogen.

(C) There is repulsion between paired electrons in oxygen's $2p$ orbitals.

(D) There is attraction between paired electrons in oxygen's $2p$ orbitals.

5. Which of the following best helps to account for the fact that the F^- ion is smaller than the O^{2-} ion?

(A) F^- has a larger nuclear mass than O^{2-} has.

(B) F^- has a larger nuclear charge than O^{2-} has.

(C) F^- has more electrons than O^{2-} has.

(D) F^- is more electronegative than O^{2-} is.

(E) F^- is more polarizable than O^{2-} is.

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6.

$$X(g) \rightarrow X^{+}(g) + e^{-} \qquad IE_1 = 740 \text{ kJ/mol}$$
$$X^{+}(g) \rightarrow X^{2+}(g) + e^{-} \qquad IE_2 = 1450 \text{ kJ/mol}$$
$$X^{2+}(g) \rightarrow X^{3+}(g) + e^{-} \qquad IE_3 = 7730 \text{ kJ/mol}$$

For element X represented above, which of the following is the most likely explanation for the large difference between the second and third ionization energies?

(A) The effective nuclear charge decreases with successive ionizations.

(B) The shielding of outer electrons increases with successive ionizations.

(C) The electron removed during the third ionization is, on average, much closer to the nucleus than the first two electrons removed were.

(D) The ionic radius increases with successive ionizations.

7. Which of the following correctly compares periodic properties of two elements and provides an accurate explanation for that difference?

(A) The first ionization energy of Al is greater than that of B because Al has a larger nuclear charge than B does.

(B) The first ionization energy of F is greater than that of O because O has a higher electronegativity than F has.

(C) The atomic radius of Ca is larger than that of Mg because the valence electrons in Mg experience more shielding than the valence electrons in Ca do.

(D) The atomic radius of Cl is smaller than that of S because Cl has a larger nuclear charge than S does.

8.

	Ionization Energies (kJ / mol)
First	590
Second	1150
Third	4910
Fourth	6490

Based on the ionization energies of element X given in the table above, which of the following is most likely the formula of the compound formed between element X and SO_4^{2-} ?

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(A) XSO_4
(B) X_2SO_4
(C) $X_2(SO_4)_3$
(D) $X(SO_4)_2$

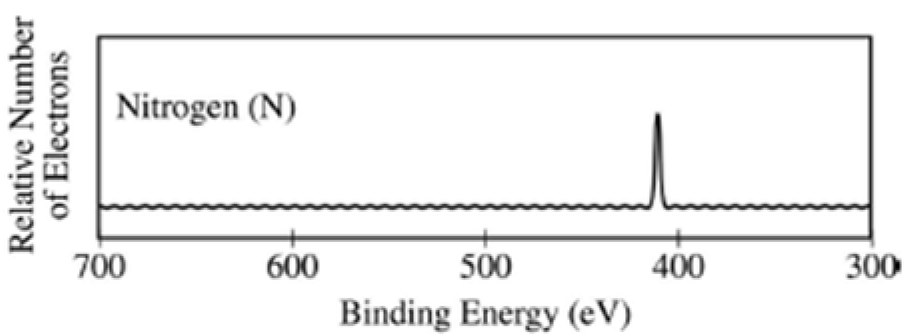
9. Which of the following best helps to explain why the electron affinity of Br has a greater magnitude than that of I?

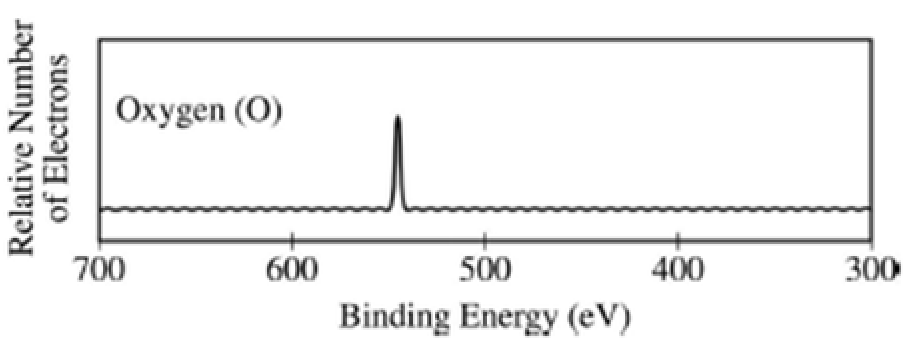
(A) Br has a lower electronegativity than I does.
(B) Br has a lower ionization energy than I does.
(C) An added electron would go into a new shell in Br but not in I.
(D) There is a greater attraction between an added electron and the nucleus in Br than in I.

10. Which of the following best helps explain why the electronegativity of Cl is less than that of F?

(A) The mass of the Cl atom is greater than the mass of the F atom.
(B) The Cl nucleus contains more protons than the F nucleus contains.
(C) When Cl and F form bonds with other atoms, the Cl bonding electrons are more shielded from the positive Cl nucleus than the F bonding electrons are shielded from the positive F nucleus.
(D) Because Cl is larger than F, the repulsions among electrons in the valence shell of Cl are less than the repulsions among electrons in the valence shell of F.

11.





The photoelectron spectra above show the energy required to remove a 1*s* electron from a nitrogen atom and from an oxygen atom. Which of the following statements best accounts for the peak in the upper spectrum being to the right of the peak in the lower spectrum?

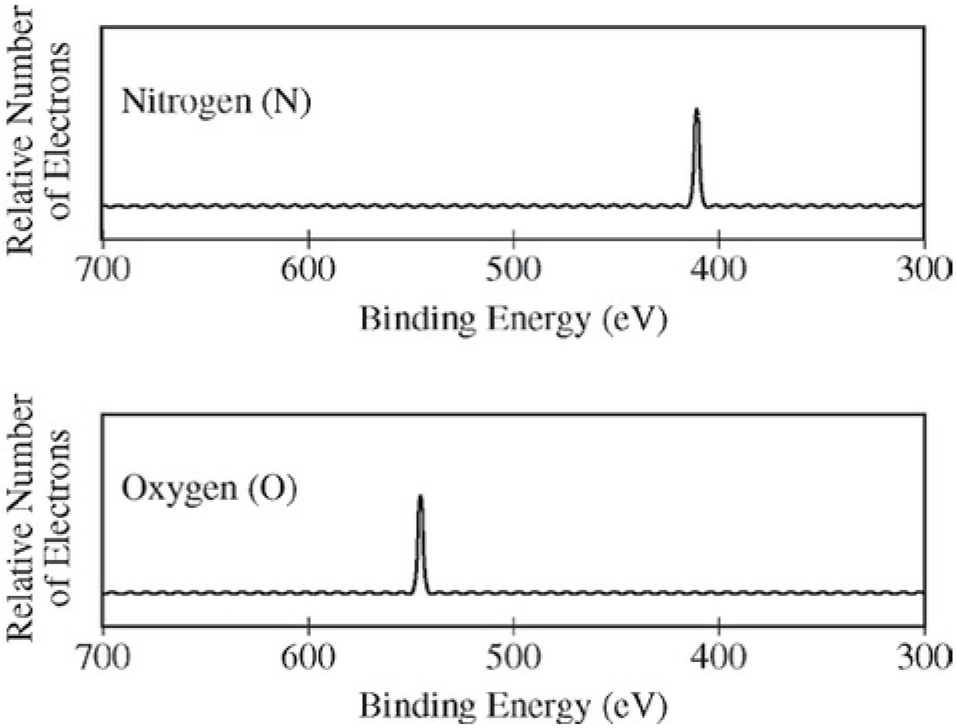
(A) Nitrogen atoms have a half-filled *p* subshell.

(B) There are more electron-electron repulsions in oxygen atoms than in nitrogen atoms.

(C) Electrons in the *p* subshell of oxygen atoms provide more shielding than electrons in the *p* subshell of nitrogen atoms.

(D) Nitrogen atoms have a smaller nuclear charge than oxygen atoms.

12.



The photoelectron spectra above show the energy required to remove a 1*s* electron from a nitrogen atom and from an oxygen atom. Which of the following statements best accounts for the peak in the upper spectrum being to the right of the peak in the lower spectrum?

(A) Nitrogen atoms have a half-filled *p* subshell.

(B) There are more electron-electron repulsions in oxygen atoms than in nitrogen atoms.

(C) Electrons in the *p* subshell of oxygen atoms provide more shielding than electrons in the *p* subshell of nitrogen atoms.

(D) Nitrogen atoms have a smaller nuclear charge than oxygen atoms.

Directions: Each set of lettered choices below refers to the numbered statements immediately following it. Select the one lettered choice that best fits each statement. A choice may be used once, more than once, or not at all in each set.

(A) Cs
(B) Ag
(C) Pb
(D) Br
(E) Se

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13. Has the highest electronegativity

(A) Cs
(B) Ag
(C) Pb
(D) Br
(E) Se

14. Has the largest atomic radius

(A) Cs
(B) Ag
(C) Pb
(D) Br
(E) Se

15. Has the lowest first-ionization energy

(A) Cs
(B) Ag
(C) Pb
(D) Br
(E) Se

---

16. The elements in which of the following have most nearly the same atomic radius?

(A) Be, B, C, N
(B) Ne, Ar, Kr, Xe
(C) Mg, Ca, Sr, Ba
(D) C, P, Se, I
(E) Cr, Mn, Fe, Co

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17.

	Ionization Energy (kJ/mol)
First	801
Second	2,430
Third	3,660
Fourth	25,000
Fifth	32,820

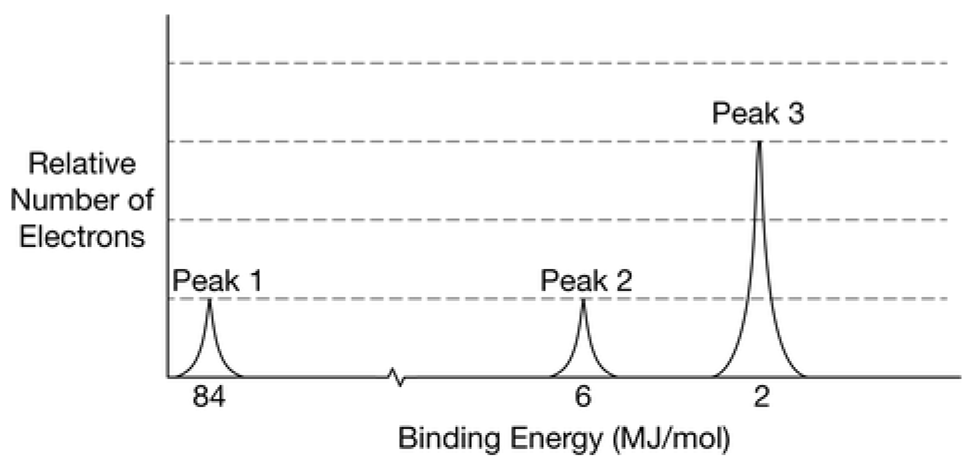
The first five ionization energies of a second-period element are listed in the table above. Which of the following correctly identifies the element and best explains the data in the table?

(A) B, because it has five core electrons

(B) B, because it has three valence electrons

(C) N, because it has five valence electrons

(D) N, because it has three electrons in the p sublevel



The complete photoelectron spectrum of a pure element is shown in the diagram above.

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18. According to the complete photoelectron spectrum, which of the following is the identity of the element?

(A) Be
(B) C
(C) O
(D) Ne

19. If peaks 1 and 2 represent s sublevels, which of the following sublevels is represented by peak 3?

(A) s
(B) p
(C) d
(D) f

20.

	Ionization Energy $(\mathrm{kJ/mol})$
First	578
Second	$1{,}817$
Third	$2{,}745$
Fourth	$11{,}577$
Fifth	$14{,}842$

The first five ionization energies of an unknown element are listed in the table above. Which of the following statements correctly identifies the element and cites the evidence supporting the identification?

(A) Na, because of the large difference between the first and the second ionization energies
(B) Al, because of the large difference between the third and fourth ionization energies
(C) Si, because the fifth ionization energy has the greatest value
(D) P, because a neutral atom of P has five valence electrons

21. Which of the following represents an electron configuration that corresponds to the valence electrons of an element for which there is an especially large jump between the second and third ionization energies? (Note: n represents a principal quantum number equal to or greater than 2.)

(A) ns^2
(B) ns^2np^1
(C) ns^2np^2
(D) ns^2np^3

22. Which of the following correctly identifies which has the higher first-ionization energy, Cl or Ar, and supplies the best justification?

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(A) Cl, because of its higher electronegativity
(B) Cl, because of its higher electron affinity
(C) Ar, because of its completely filled valence shell
(D) Ar, because of its higher effective nuclear charge

23.

Ionization Energies for element X (kJ mol^{-1})				
First	Second	Third	Fourth	Fifth
580	1,815	2,740	11,600	14,800

The ionization energies for element *X* are listed in the table above. On the basis of the data, element *X* is most likely to be

(A) Na
(B) Mg
(C) Al
(D) Si
(E) P

24. Which of the following lists Mg, P, and Cl in order of increasing atomic radius?

(A) Cl < P < Mg
(B) Cl < Mg < P
(C) Mg < P < Cl
(D) Mg < Cl < P
(E) P < Cl < Mg

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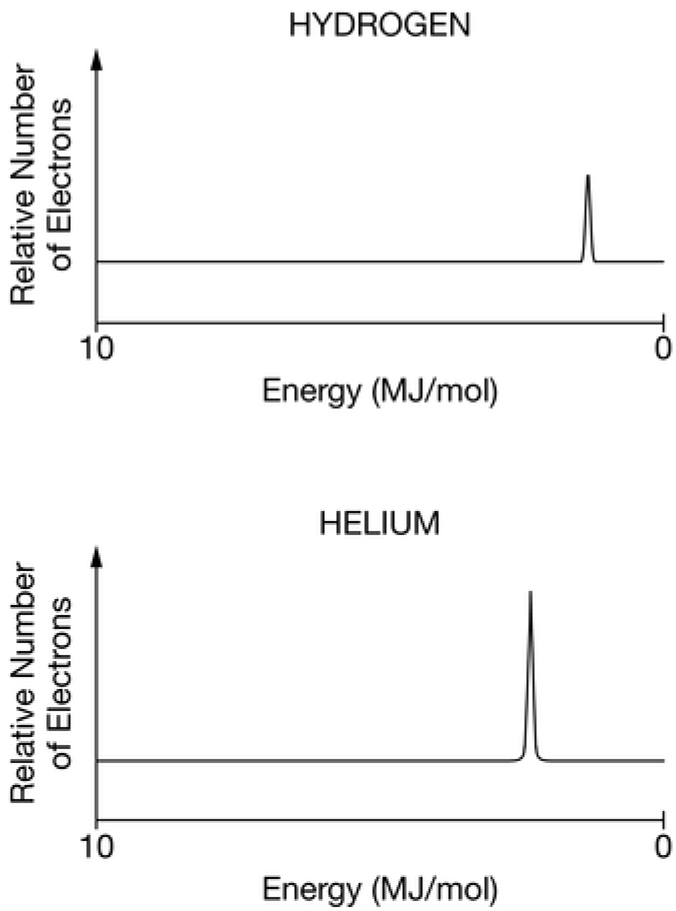
25.



The photoelectron spectra for H and He are represented above. Which of the following statements best accounts for the fact that the peak on the He spectrum is farther to the left and higher than the peak on the H spectrum?

(A) He has an additional valence electron in a higher energy level than the valence electron in H.

(B) He has a greater nuclear charge than H and an additional electron in the same energy level.

(C) He has a completely filled valence shell in which the electrons are a greater distance from the nucleus than the distance between the H nucleus and its electron.

(D) It takes longer for the electrons in He to be removed due to the higher nuclear mass of He.

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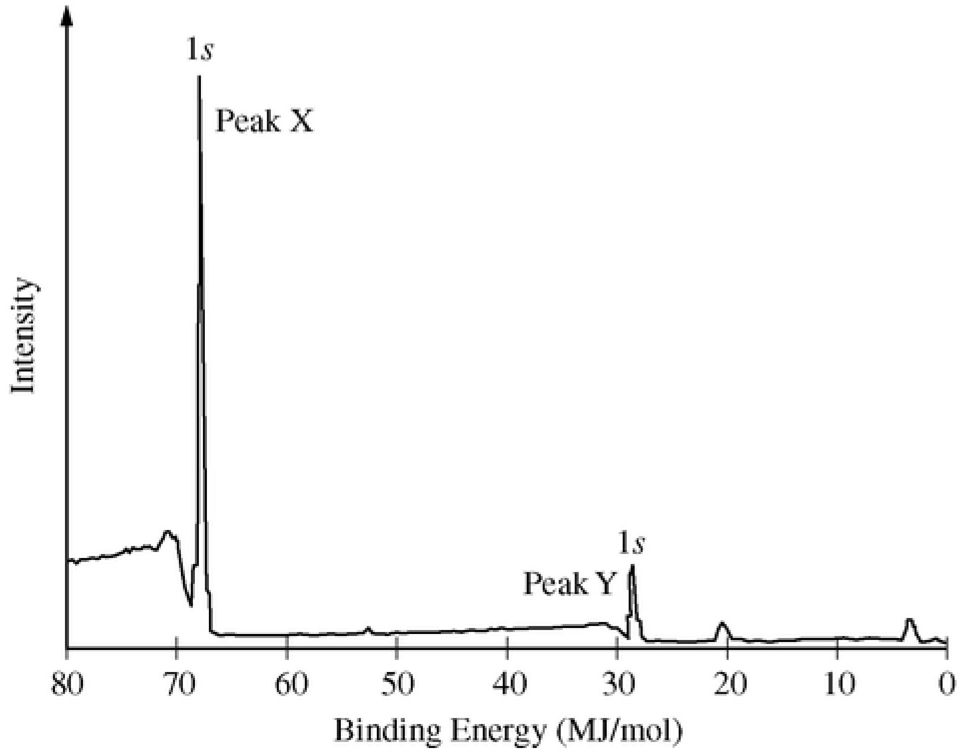
26.



A sample containing atoms of C and F was analyzed using x-ray photoelectron spectroscopy. The portion of the spectrum showing the $1s$ peaks for atoms of the two elements is shown above. Which of the following correctly identifies the $1s$ peak for the F atoms and provides an appropriate explanation?

(A) Peak X, because F has a smaller first ionization energy than C has.

(B) Peak X, because F has a greater nuclear charge than C has.

(C) Peak Y, because F is more electronegative than C is.

(D) Peak Y, because F has a smaller atomic radius than C has.

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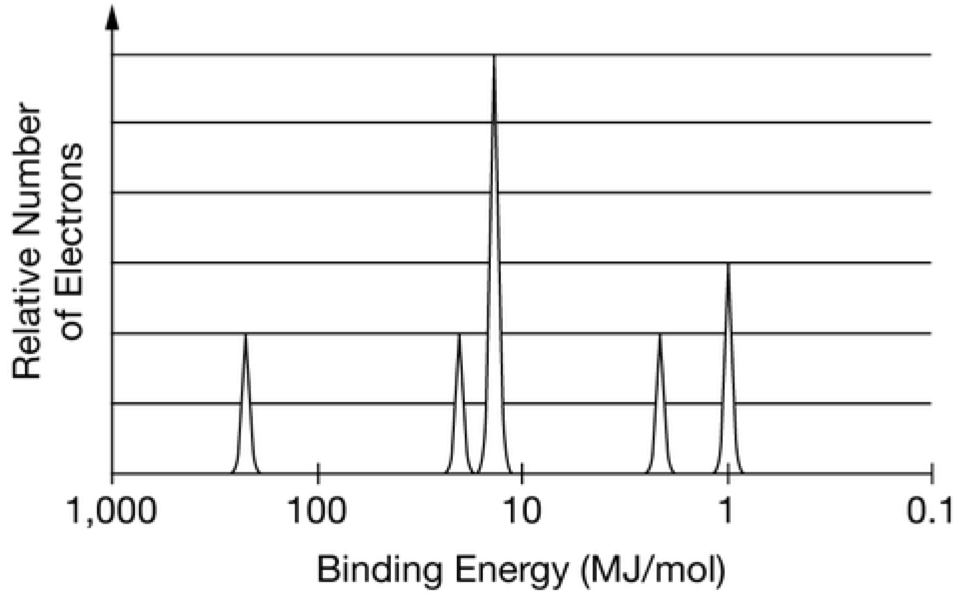
27.



The complete photoelectron spectrum of an element is given above. Which of the following electron configurations is consistent with the spectrum?

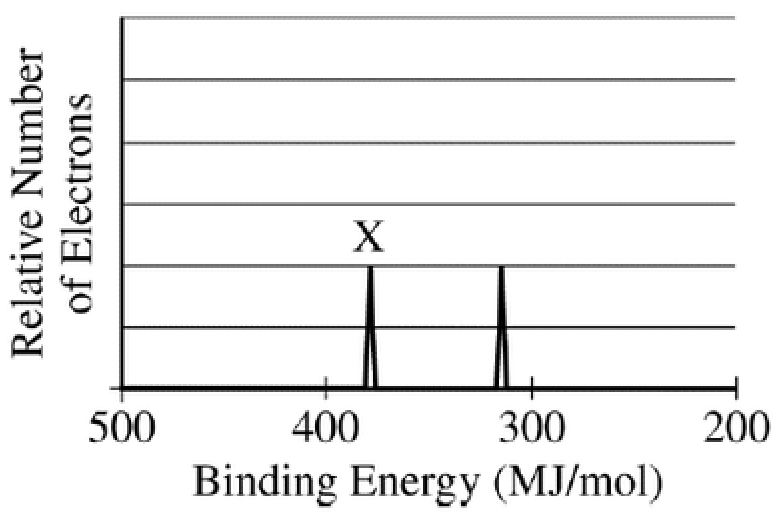
(A) $1s^2\ 2s^2 2p^1$

(B) $1s^2\ 2s^2 2p^6\ 3s^2 3p^3$

(C) $1s^2\ 2s^2 2p^6\ 3s^2 3p^6$

(D) $1s^2\ 2s^2 2p^6\ 3s^2 3p^6 4s^2\ 3d^5$

28.



The photoelectron spectra of the $1s$ electrons of two isoelectronic species, Ca^{2+} and Ar, are shown above. Which of the following correctly identifies the species associated with peak X and provides a valid justification?

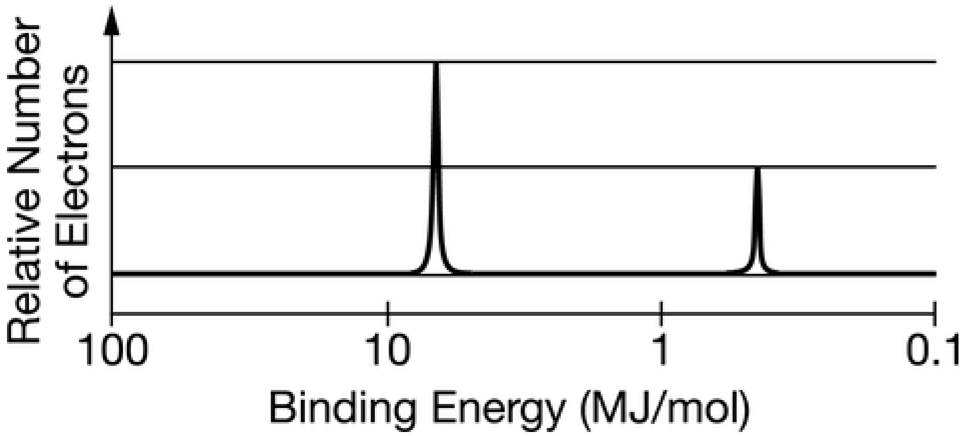
(A) Ar, because it has completely filled energy levels

(B) Ar, because its radius is smaller than the radius of Ca^{2+}

(C) Ca^{2+}, because its nuclear mass is greater than that of Ar

(D) Ca^{2+}, because its nucleus has two more protons than the nucleus of Ar has

29.



The complete photoelectron spectrum for an element is shown above. Which of the following observations would provide evidence that the spectrum is consistent with the atomic model of the element?

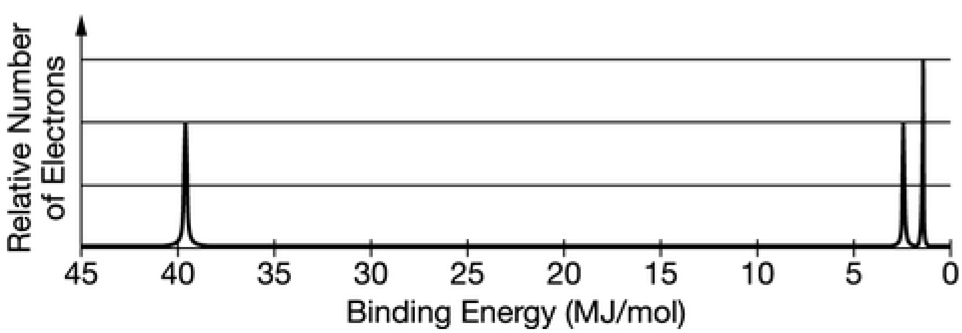
(A) A neutral atom of the element contains exactly two electrons.

(B) The element does not react with other elements to form compounds.

(C) In its compounds, the element tends to form ions with a charge of $+1$.

(D) In its compounds, the element tends to form ions with a charge of $+3$.

30.



The photoelectron spectrum for the element nitrogen is represented above. Which of the following best explains how the spectrum is consistent with the electron shell model of the atom?

(A) The leftmost peak represents the valence electrons.

(B) The two peaks at the right represent a total of three electrons.

(C) The electrons in the $1s$ sublevel have the smallest binding energy.

(D) The electrons in the $2p$ sublevel have the smallest binding energy.

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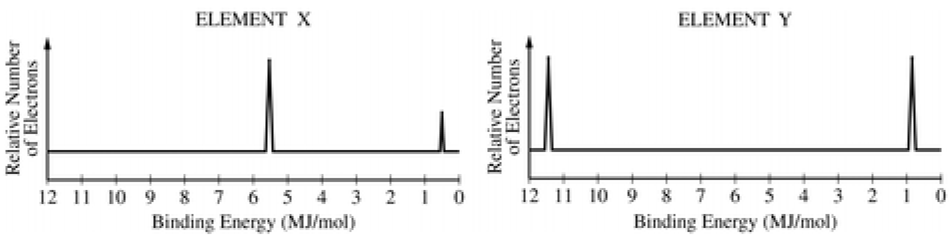
31.



The complete photoelectron spectra of neutral atoms of two unknown elements, X and Y, are shown above. Which of the following can be inferred from the data?

(A) Element X has a greater electronegativity than element Y does.

(B) Element X has a greater ionization energy than element Y does.

(C) Element Y has a greater nuclear charge than element X does.

(D) The isotopes of element Y are approximately equal in abundance, but those of element X are not.

32.

Element	Atomic Radius	First Ionization Energy
Calcium	194 pm	590 kJ/mol
Potassium	—	—

Based on periodic trends and the data in the table above, which of the following are the most probable values of the atomic radius and the first ionization energy for potassium, respectively?

(A) 242 pm, 633 kJ/mol

(B) 242 pm, 419 kJ/mol

(C) 120 pm, 633 kJ/mol

(D) 120 pm, 419 kJ/mol

33. Which of the following correctly compares the atomic radius of F to that of O and provides the best explanation?

(A) The atomic radius of F is larger than that of O because F has more occupied subshells than O does.

(B) The atomic radius of F is larger than that of O because F has more protons in the nucleus than Odoes.

(C) The atomic radius of F is smaller than that of O because F has a greater effective nuclear charge than O does.

(D) The atomic radius of F is smaller than that of O because valence electrons in F experience more shielding than those in O.

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34.

Element	Second Ionization Energy (kJ/mol)
Li	7300
Na	4560
K	3050
Rb	2630

The decrease in the second ionization energy of alkali metals going down the group, as shown in the table above, can be best attributed to a decrease in the coulombic force of attraction due to

(A) the removal of core electrons
(B) an increase in effective nuclear charge
(C) an increase in the average distance of the outermost electron from the nucleus
(D) a decrease in electron-electron repulsion within the outermost shell

35. Which of the following properties generally decreases across the periodic table from sodium to chlorine?

(A) First ionization energy
(B) Atomic mass
(C) Electronegativity
(D) Maximum value of oxidation number
(E) Atomic radius

36. Which of the following elements has the largest first ionization energy?

(A) Li
(B) Be
(C) B
(D) C
(E) N

37. Atoms of Mg combine with atoms of F to form a compound. Atoms of which of the following elements combine with atoms of F in the same ratio?

(A) Li
(B) Ba
(C) Al
(D) Cl
(E) Ne

When free $Cl(g)$ atoms encounter $O_3(g)$ molecules in the upper atmosphere, the following reaction mechanism is proposed to occur.

$Cl(g) + O_3(g) \rightarrow ClO(g) + O_2(g)$	*slow step*	
$ClO(g) + O_3(g) \rightarrow Cl(g) + 2\,O_2(g)$	*fast step*	
$2\,O_3(g) \rightarrow 3\,O_2(g)$	*overall reaction*	$\Delta H = -285$ kJ/mol$_{rxn}$

38.

$X + O_3 \rightarrow XO + O_2$
$XO + O_3 \rightarrow X + 2\,O_2$
————————
$2\,O_3 \rightarrow 3\,O_2$

The proposed mechanism can be written in a more general form, as shown above. Species other than Cl can also decompose O_3 through the same mechanism. Which of the following chemical species is most likely to decompose O_3 in the upper atmosphere through the above mechanism?

(A) He

(B) Br

(C) N_2

(D) O_2

39. Based on the positions of elements in the periodic table, which of the following groups contains compounds that are most likely to have similar chemical properties?

(A) KCl, KBr, and KI

(B) $AsCl_3$, SCl_2, and ClF

(C) CF_4, PF_3, and SeF_2

(D) NH_3, H_2O, and HF

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40.

	Ionization Energy (kJ/mol)
First	577
Second	1,816
Third	2,745
Fourth	11,577
Fifth	14,482

Based on the ionization energies of element X given in the table above, which of the following is most likely the empirical formula of an oxide of element X?

(A) XO_2
(B) X_2O
(C) X_2O_3
(D) X_2O_5

41.

	Ionization Energy (kJ/mol)
First	730
Second	1450
Third	7700
Fourth	10,500

The ionization energies of an unknown element, X, are listed in the table above. Which of the following is the most likely empirical formula of a compound formed from element X and phosphorus, P ?

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(A) XP
(B) X_3P
(C) X_3P_2
(D) X_3P_4

42. $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^3$

Atoms of an element, X, have the electronic configuration shown above. The compound most likely formed with magnesium, Mg, is

(A) MgX
(B) Mg_2X
(C) MgX_2
(D) MgX_3
(E) Mg_3X_2

43. The compound CCl_4 is nonflammable and was once commonly used in fire extinguishers. On the basis of the periodic properties, which of the following compounds can most likely be used as a fire-resistant chemical?

(A) BCl_3
(B) CH_4
(C) CBr_4
(D) $PbCl_2$

44. Forms monatomic ions with 2– charge in solutions

(A) F
(B) S
(C) Mg
(D) Ar
(E) Mn

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45.

Element	Known Oxides
H	H_2O, H_2O_2
Li	Li_2O, Li_2O_2
Na	Na_2O, Na_2O_2, NaO_2
K	K_2O, K_2O_2, KO_2

Based on the information above and periodic trends, which of the following is the best hypothesis regarding the oxide(s) formed by Rb?

(A) Rb will form only Rb_2O.

(B) Rb will form only Rb_2O_2.

(C) Rb will form only Rb_2O and Rb_2O_2.

(D) Rb will form Rb_2O, Rb_2O_2, and RbO_2.

46. If Na reacts with chlorine to form NaCl, which of the following elements reacts with Na to form an ionic compound in a one-to-one ratio, and why?

(A) K, because it is in the same group as Na.

(B) Mg, because its mass is similar to that of Na.

(C) Ar, because its mass is similar to that of Cl.

(D) Br, because it has the same number of valence electrons as Cl.

47.

Metal	Be	Mg	Ca	Sr	Ba	Ra
Formula of Metal Chloride	$BeCl_2$	$MgCl_2$	$CaCl_2$	$SrCl_2$	$BaCl_2$	$RaCl_2$

All the chlorides of the alkaline earth metals have similar empirical formulas, as shown in the table above. Which of the following best helps to explain this observation?

(A) $Cl_2(g)$ reacts with metal atoms to form strong, covalent double bonds.

(B) Cl has a much greater electronegativity than any of the alkaline earth metals.

(C) The two valence electrons of alkaline earth metal atoms are relatively easy to remove.

(D) The radii of atoms of alkaline earth metals increase moving down the group from Be to Ra.

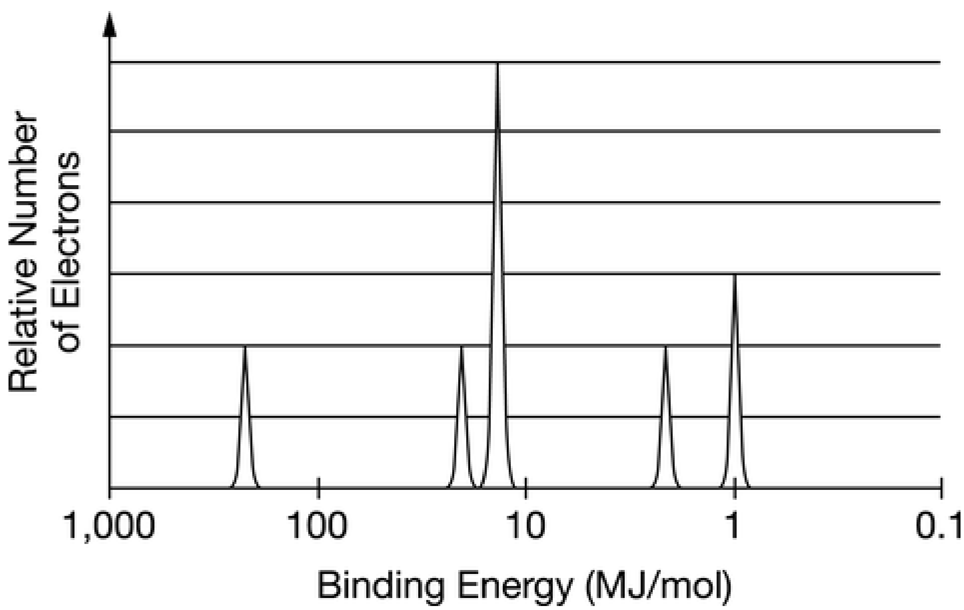
48. RbCl has a high boiling point. Which of the following compounds is also likely to have a high boiling point, and why?

(A) NO, because its elements are in the same period of the periodic table.
(B) ClF, because its elements are in the same group of the periodic table.
(C) Cl_2O, because its elements have similar electronegativities and it is a covalent compound.
(D) $CsCl$, because its elements have very different electronegativities and it is an ionic compound.

49. Which of the following best helps to explain why $Na(s)$ is more reactive with water than $Mg(s)$ is?

(A) $Na(s)$ is softer than $Mg(s)$.
(B) The atomic mass of Na is less than that of Mg.
(C) The Na^{+} ion has weaker Coulombic attraction to an anion than the Mg^{2+} ion has.
(D) The first ionization energy of Na is less than that of Mg.

50. For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.



The photoelectron spectrum for an unknown element is shown above.

(a) Based on the photoelectron spectrum, identify the unknown element and write its electron configuration.

(b) Consider the element in the periodic table that is directly to the right of the element identified in part (a). Would the $1s$ peak of this element appear to the left of, the right of, or in the same position as the $1s$ peak of the element in part (a)? Explain your reasoning.

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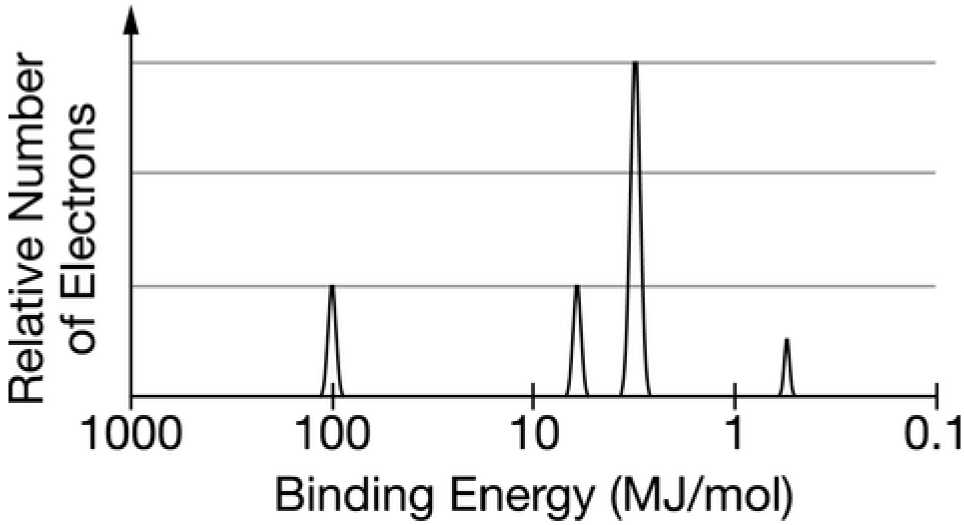
51. For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.



The complete photoelectron spectrum of an unknown element is given above.

(a) Draw an X above the peak that corresponds to the orbital with electrons that are, on average, closest to the nucleus. Justify your answer in terms of Coulomb’s law.

(b) Based on the spectrum, write the complete electron configuration of the element.

(c) On the graph, draw the peak(s) corresponding to the valence electrons of the element that has one more proton in its nucleus than the unknown element has.